

Course syllabus for

Academic year 2025/2026 >

MCCX04 - Master's thesis in Microtechnology and nanoscience

Masterexamensarbete vid Mikroteknologi och nanovetenskap

Course syllabus adopted 2024-01-30 by Vice President (or corresponding)

Owner: [TKELT](#)

30,0 Credits

Grading: UG - Pass, Fail

Education cycle: Second-cycle

Main field of study: Electrical Engineering, Engineering Physics

Department: 59 - MICROTECHNOLOGY AND NANOSCIENCE

Course round 1

Teaching language: English

Application code: 50123

Open for exchange students: No

Credit distribution

Module	Sp1	Sp2	Sp3	Sp4	Summer course	No Sp	Examination dates
0108 Diploma thesis 30,0 c Grading: UG	15,0 c	15,0 c					

Examiner:

[Per Rudquist](#)

[Go to course homepage](#)

Course round 2

Teaching language: English

Application code: 50149

Open for exchange students: No

Credit distribution

Module	Sp1	Sp2	Sp3	Sp4	Summer course	No Sp	Examination dates
0108 Diploma thesis 30,0 c Grading: UG			15,0 c	15,0 c			

In programs

[MPEES Embedded Electronic System Design, Year 2](#)

[MPNAT Nanotechnology, Year 2](#)

[MPPHS Physics, Year 2](#)

[MPWPS Wireless, Photonics and Space Engineering, Year 2](#)

Examiner:

[Per Rudquist](#)

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Eligibility

General entry requirements for Master's level (second cycle)

Specific entry requirements

English 6 (or by other approved means with the equivalent proficiency level)

Course specific prerequisites

- Students pursuing the five-year Master of Science in Engineering/Master of Architecture education must have accrued at least 225 hec before beginning work on a thesis.
- Students who are only taking part in the Master's programme must have accrued at least 45 hec in this programme before beginning work on a thesis.
- In addition to the general credit requirements, the necessary prerequisite courses must be completed for

the specific thesis. The examiner will formulate and monitor these prerequisite requirements.

Aim

The purpose of a thesis is to enable the student to develop deeper knowledge, understanding, capabilities and attitudes in the context of the programme of study. The thesis should be written at the end of the programme and offers the opportunity to delve more deeply into and synthesise knowledge acquired in previous studies. A thesis for a Master of Science in Engineering/Master of Architecture or Master of Science programmes should place emphasis on the technical/scientific/architectural aspects of the subject matter.

Learning outcomes (*after completion of the course the student should be able to*)

The learning objectives for a thesis are based on the objectives for Master of Science in Engineering/Architecture/Master of Science degrees formulated in Chalmers' local Master's degree procedures.

Specific learning objectives that are to be fulfilled in the master's thesis are that the student should be able to:

- in their project use essential in-depth knowledge of the major subject/field of study and, in a scientifically correct way, relate to current research and development work;
- choose and state one's reasons for selecting their project method with respect to the major subject/field of study;
- contribute to research and development work, and be able to relate their work to relevant scientific or technical/industrial/architectonic contexts;
- based on a holistic view, critically, independently and creatively identify, formulate and deal with complex issues;
- plan and, with adequate methods, carry out qualified tasks within the designated framework and also be able to evaluate this work;
- create, analyse and critically evaluate different technical/architectonic solutions;
- critically and systematically integrate knowledge;
- present and discuss their conclusions as well as the knowledge and arguments these are based upon both in spoken and written English;
- within the framework for the specific project, identify which issues need to be addressed for relevant societal, ethical and ecological factors to be observed, and
- observe and discuss ethical aspects of research and development work, both pertaining to how the work is carried out as well as what it explores/develops;
- identify and discuss needs for further elucidation of different project aspects before decision-making or project realisation, when relevant.

Upon completion of the degree work, the student shall have demonstrated such knowledge and ability as are required to work independently as a Master of Architecture/Master of Science in Engineering/ Master of Science.

Content

The task consists of independent work, which can be carried out individually or in pairs. The task includes topic selection, problem formulation and problem definition, literature studies, empirical data collection including practical experiments, development, preparation of a report as well as presentation and defense of the report at a seminar. The report must meet certain scientific criteria. The course also includes opposition on another group's thesis report.

Organisation

Initiation

- The thesis is to be written by a single student or two students working together.
- The student contacts an examiner or the master program responsible can suggest an examiner.
- The student formulates a brief written description of the topic for the thesis. The description should contain the background, purpose, objective and possibly the method.
- The examiner will ensure that the student and the project meet the requirements for general and specific prerequisites and the general intended learning outcomes for theses.
- The master program responsible ensures that the thesis falls within the main subject of the Master's programme. If the student is accepted in a five years program for Master of Science in Engineering or Architecture, the master program responsible should as well assess and approve that the master thesis is relevant for the appropriate technical/architectural subject area.
- The student submits information of the thesis project via the web form at Chalmers' education pages, Your studies. The examiner and the master program responsible sign their approval before registration in Ladok is done.

Planning

- The student must write a planning report providing a detailed description of the problem/task. The planning report must contain the background, purpose, objective, scope, method and timetable for completion of the thesis. The planning report is submitted to the examiner for approval.

Supervision

- The students are entitled to guidance during the course of their work.

Written report

- The thesis must be written in English.
- The cover, etc., of the thesis must be formulated according to the instruction and templates at "Design and Publish master Thesis" at Chalmers. See Chalmers Education pages, Your studies.
- The thesis must be parsed using anti-plagiarism software.
- Chalmers' policies regarding open access and confidentiality apply to the work on the thesis and to publishing the report and the master thesis in whole should be open access.
- The thesis must be published electronically in Chalmers Open Digital Repository (ODR) Student theses according to the applicable rules for electronic publication of theses at Chalmers.
- When two students work jointly on a thesis, the division of the work must be clearly stated in the thesis.

Oral presentation

- The oral presentation begins with the student describing the work. This is followed by a defense of the findings, and a discussion.
- The oral presentation must be made in English. This presentation may not take more than 60 minutes, of which approximately 1/3 of the time should be devoted to a defense of the thesis and discussion.
- At the time of the oral presentation, the written thesis must be completed but not published. This is to make it possible to include viewpoints that arise during the oral presentation to be incorporated into the thesis.
- The oral presentation, including a defense of the thesis, must be done at Chalmers at an open seminar. Any additional presentations can be made at a company or other external organisations, if necessary. In exceptional cases, for example, if thesis research has been conducted abroad, the examiner may grant a dispensation from the requirement for an oral presentation, defense of the findings in the thesis and presence at other presentations at Chalmers.
- The presentation of the thesis must be announced at the department at least two weeks prior to the date of the presentation. The presentation must be held in the period between August 15 and June 15, during normal office hours.

Additional information on the execution of the master thesis work can be available at the department.

Examination including compulsory elements

For a thesis to be approved, the following stages must be completed:

- An approved planning report
- An approved thesis
- An approved presentation and defense of the thesis
- Approved opposition of another thesis
- Attendance of the presentation and defense of two other theses

The course examiner may assess individual students in other ways than what is stated above if there are special reasons for doing so, for example if a student has a decision from Chalmers about disability study support.